Week-8:

1. Which of the following best describes Pixel Attribution Methods?
    a. They compress neural network layers to reduce model size.
    b. They automatically remove backdoor triggers from poisoned datasets.
    c. They convert text datasets directly into image representations.
    d. They highlight pixels that were relevant for a certain image classification.

Explanation: Interpretability - I [3:40-4:55] - Pixel Attribution Methods highlight the pixels that were relevant for a certain image classification by a neural network

2. Which of the following pixel attribution methods involve manipulating parts of an image to generate explanations?
    a. Gradient-based methods
    b. Occlusion-based methods
    c. Mechanical methods
    d. Top-down methods

Explanation: Interpretability - I [6:33-7:25] - Occlusion- or perturbation-based: Methods like SHAP and LIME manipulate parts of the image to generate explanations (model-agnostic).

3. What should happen to a saliency map if a model that genuinely captures higher-level class concepts has its layers randomized?
    a. The map should invert its colors.
    b. The map should remain identical.
    c. The map should change significantly.
    d. The map should become a blank image.

Explanation: Interpretability - I [20:11-22:28] - If a model captures higher-level class concepts, then saliency maps should change as the model is being randomized.

4. How can we validate the explanations given by LIME? **(Select all that apply.)**
    a. By using prior knowledge.
    b. By randomizing the model's layers.
    c. By using human annotators.
    d. By applying adversarial perturbations.

Explanation: Interpretability - I [25:10-27:03] - Using prior knowledge, we can validate the explanations. Human annotation can be used to decide whether the model's explanations are correct.

5. Which of the following interpretability approaches involves locating information in a model without fully understanding how it is processed?
    a. Mechanistic interpretability
    b. Top-down interpretability
    c. Feature-based visualization
    d. Game-theoretic approach

Explanation: Interpretability - II [9:57-11:58] - Top-down locates information in a model without full understanding of how it is processed

6. What is the primary function of probing in interpretability?
    a. To inject hidden trojans into public datasets during training.
    b. To check whether representations encode information reliably
    c. To optimize latent space interpolation for image generation
    d. To completely reverse-engineer all node-to-node algorithmic connections in large language models.

    Explanation: Interpretability - II [13:57-16:33] - Probing loosely refers to a class of methodologies in interpretability to check whether representations encode information reliably for some specific task.

7. According to the lecture, which of the following are Attack Vectors? **(Select all that apply.)**
    a. API keys
    b. Public Datasets
    c. HuggingFace models
    d. A downloadable execution file

    Explanation: Interpretability - II [20:32-21:23] - Public datasets and model sharing libraries are some examples of attack vectors.

8. Consider the following equation for LIME:

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} \quad \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

    What do the components L and $\Omega(g)$ represent?
    a. L measures how unfaithful the explanation model g is in approximating f locally (weighted by π_x, while $\Omega(g)$ measures the complexity/interpretability penalty of the explanation model.
    b. L represents the global accuracy of the black-box model f, and $\Omega(g)$ represents the total number of convolutional layers.
    c. L is the adversarial perturbation budget, and $\Omega(g)$ is the learning rate of the surrogate model.
    d. L calculates the exact Shapley values, and $\Omega(g)$ computes the gradient with respect to input pixels.

    Explanation: Interpretability - I [32:03-34:57] - In LIME's objective function, L quantifies how poorly the simple explanation model g approximates the complex black-box model f within the local neighborhood defined by π_x, while $\Omega(g)$ penalizes model complexity to ensure the resulting explanation remains simple and interpretable for humans.

9. What is the primary goal of Neural Cleanse when detecting Trojans in neural networks?
    a. To fine-tune the model using a 0.05% poisoned dataset
    b. To reverse-engineer potential triggers

c. To prune the entire first-layer weights of a classifier
d. To train a neural network using high-dimensional pooling layers

Explanation: Interpretability - II [26:57-31:36] - If we know the general form of the attack, we can reverse-engineer the Trojan by searching for a trigger and target label.

10. How does top-down interpretability differ from mechanistic interpretability?
    a. Top-down focuses on algorithmic node-to-node connections, whereas mechanistic looks at global activity of populations of neurons.
    b. Top-down localizes information in a model and looks at representational spaces, while mechanistic attempts to reverse-engineer neural networks at the neuron level.
    c. Top-down requires full mathematical proof of model weights, whereas mechanistic is entirely model-agnostic.
    d. Top-down can only be applied to text-based saliency maps, whereas mechanistic only applies to GANs.

Explanation: Interpretability - II [6:27-11:58] - Mechanistic: Reverse-engineer neural Networks; Top-Down: Locate information in a model without full understanding of how it is processed.

11. What game-theoretic concept does SHAP (Shapley Additive exPlanations) explain about the output of any machine learning model?
    a. Nash Equilibrium
    b. Pareto Optimality
    c. Shapley Values
    d. Minimax Theorem

Explanation: Interpretability - I [35:20-39:39] - Connects optimal credit allocation with local explanations using the classic Shapley values from game theory and their related extensions.

12. Regarding LIME and SHAP, which of the following statements are true? **(Select all that apply.)**
    a. They generate global explanations that apply uniformly across the entire dataset.
    b. They are model-agnostic methods that can work on any black-box model.
    c. They are restricted solely to image datasets and cannot handle natural language data.
    d. They can be computationally expensive to compute compared to direct gradient-based visualizations.

Explanation: Interpretability - I [25:09-39:39] - Works on any black-box model with many data types.